

Note 3.6 The Chain Rule and Derivatives of Inverse Functions

The Chain Rule

- **Definition:**

$y = f(g(x))$ is composite function of $y = f(u)$, $u = g(x)$, then

$$(f(g(x)))' = f'(g(x)) \cdot g'(x)$$

or

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

Remarks: $y = f(u)$, $u = g(v)$, $v = h(x)$, then

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx} = f'(u) \cdot g'(v) \cdot h'(x)$$

Derivative of Inverse Functions

- **Definition:**

If $f'(x)$ exists and never zero on domain and f^{-1} is differentiable at every point in its domain, then at the point $a = f^{-1}(b)$,

$$(f^{-1})'(b) = \frac{1}{f'(f^{-1}(b))}$$

or

$$\left. \frac{df^{-1}}{dx} \right|_{x=b} = \frac{1}{\left. \frac{df}{dx} \right|_{x=f^{-1}(b)}}$$

- **Derivatives of some Trigonometric Functions**

- $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$

- $(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$

- $(\arctan x)' = \frac{1}{1+x^2}$

•

$$(\operatorname{arc\,cot} x)' = -\frac{1}{1+x^2}$$